

16

Wk 16 106-259

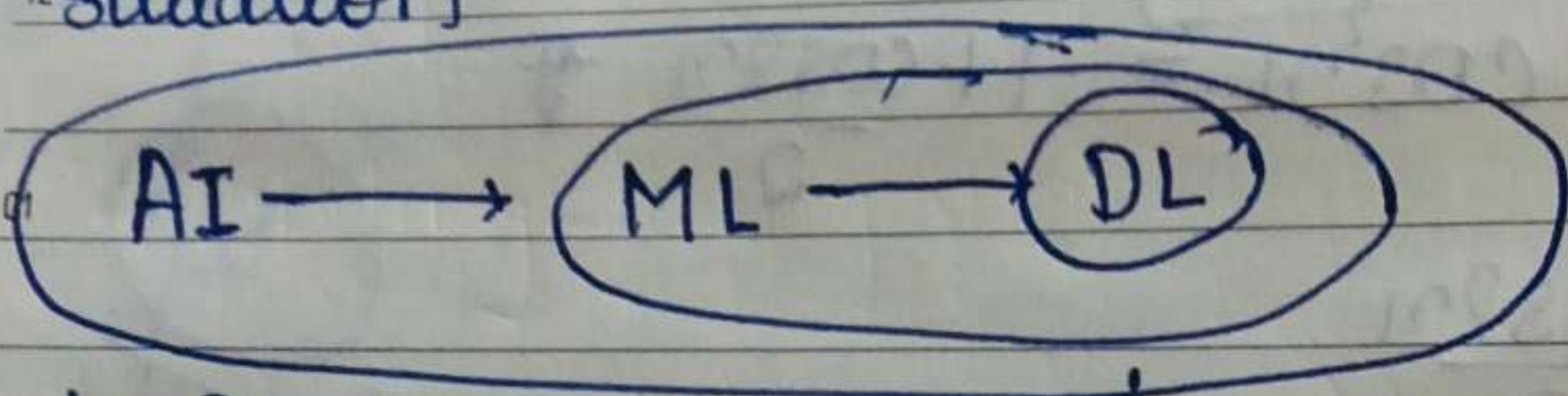
April
Monday
2018

9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29

L-1

Machine learning → System identifies patterns in data and uses those patterns to make predictions or decisions

→ It is the process of teaching computers to learn from data so they can make predictions, recognize patterns, and improve their performance over time without being explicitly programmed for every situation



L-2

ML → Imp

Supervised

- Regression
- Classification

Unsupervised

→ Clustering

→ Dimensionality reduction

→ Anomaly detection

→ Association Rule learning

Semi-supervised

Reinforcement

Supervised → Data have both input / Output have to predict on the basis of that.

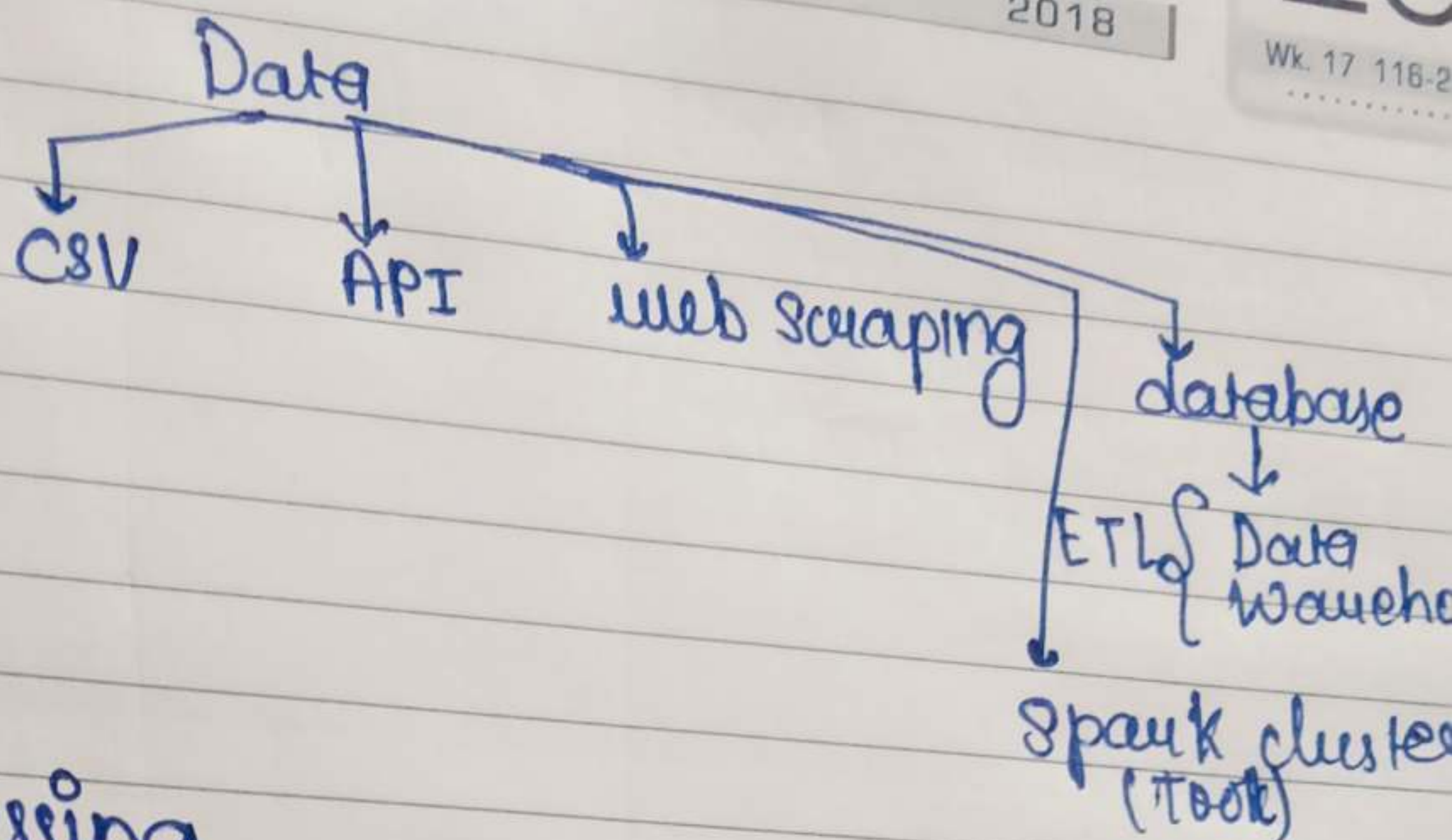
14 15 16 17 18 19 20
21 22 23 24 25 26 27
28 29 30 31

(MLDLC/MLDC).
Machine Learning Development Life Cycle

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2) Gathering data



3) Data Preprocessing

- ↳ remove duplicate
- ↳ remove null/missing
- ↳ outliers
- ↳ Scale.

4) Exploratory Data Analysis

- ↳ Visualizations
- ↳ univariate / Bivariate
- ↳ outlier
- ↳ Imbalanced

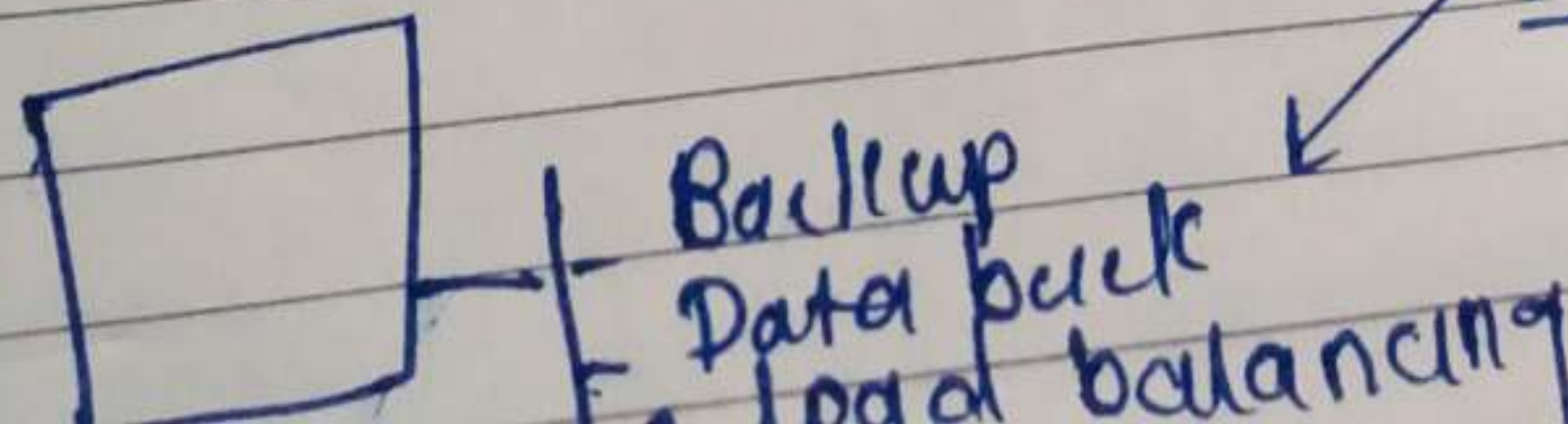
5) Feature Engineering and Selection

6) Model Training, Evaluation and Selection → Ensemble learning

7) Model Deployment

8) Testing

9) Optimise



4) Popu Quality data

5) Irrelevant Features (columns)

→ Garbage In garbage out

6) Overfitting

7) Underfitting

8) Software Integration

9) Offline Learning / Development

10) Cost Involved

Lecture - 8

Applications of Machine Learning

Retail - Amazon / Big - Bazar

Banking & Finance

Transport - OLA

Manufacturing Tesla

Consumer Internet - Twitter

2) Cather

3) Data

- ↳ remove
- ↳ remove
- ↳ outlier
- ↳ scale

4) Explora

- ↳ Visualiza
- ↳ univari
- ↳ outlier
- ↳ Imbalan

5) Feature E

6) Model Train

7) Model Deple

8) Testing

9) Optimize

08 Image Classificaⁿ
or anything related
to ML - where data
pattern remains
constant - without
sudden concept
drift

Applⁿ

Used in finance,
economics, health where
new data patterns are
constantly emerging

Industry proven
tools. Eg. Scikit,
Tensorflow, Pytorch,
Keras, Spack MLlib.

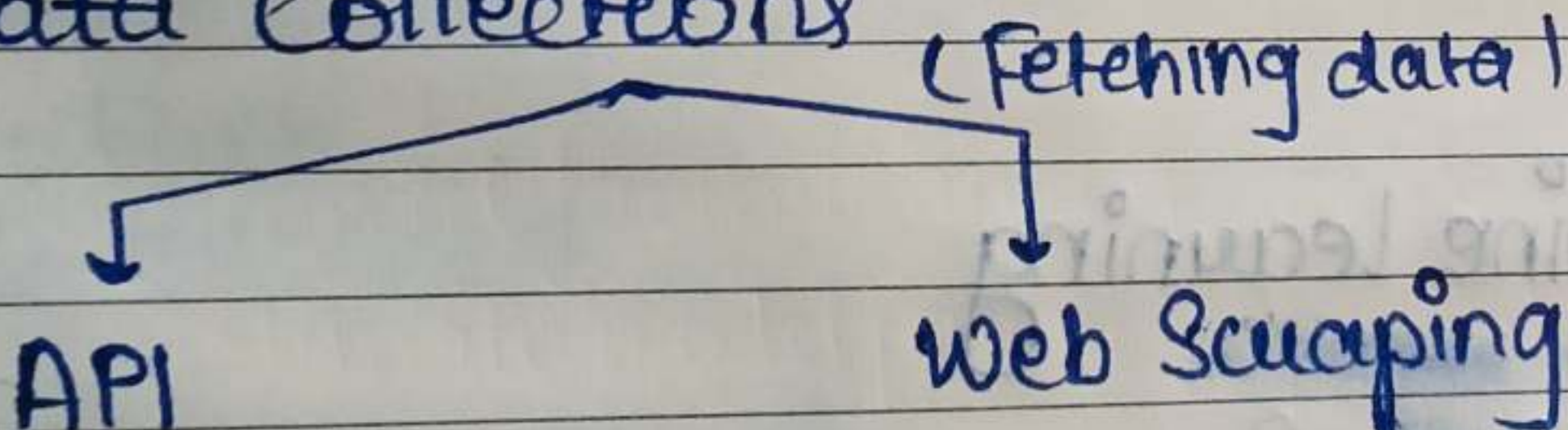
Tools

Active research / New
project tools: Eg. MOA,
SAMOA, Scikit-multiflow,
StreamDM

(L-7) (24/08/26)

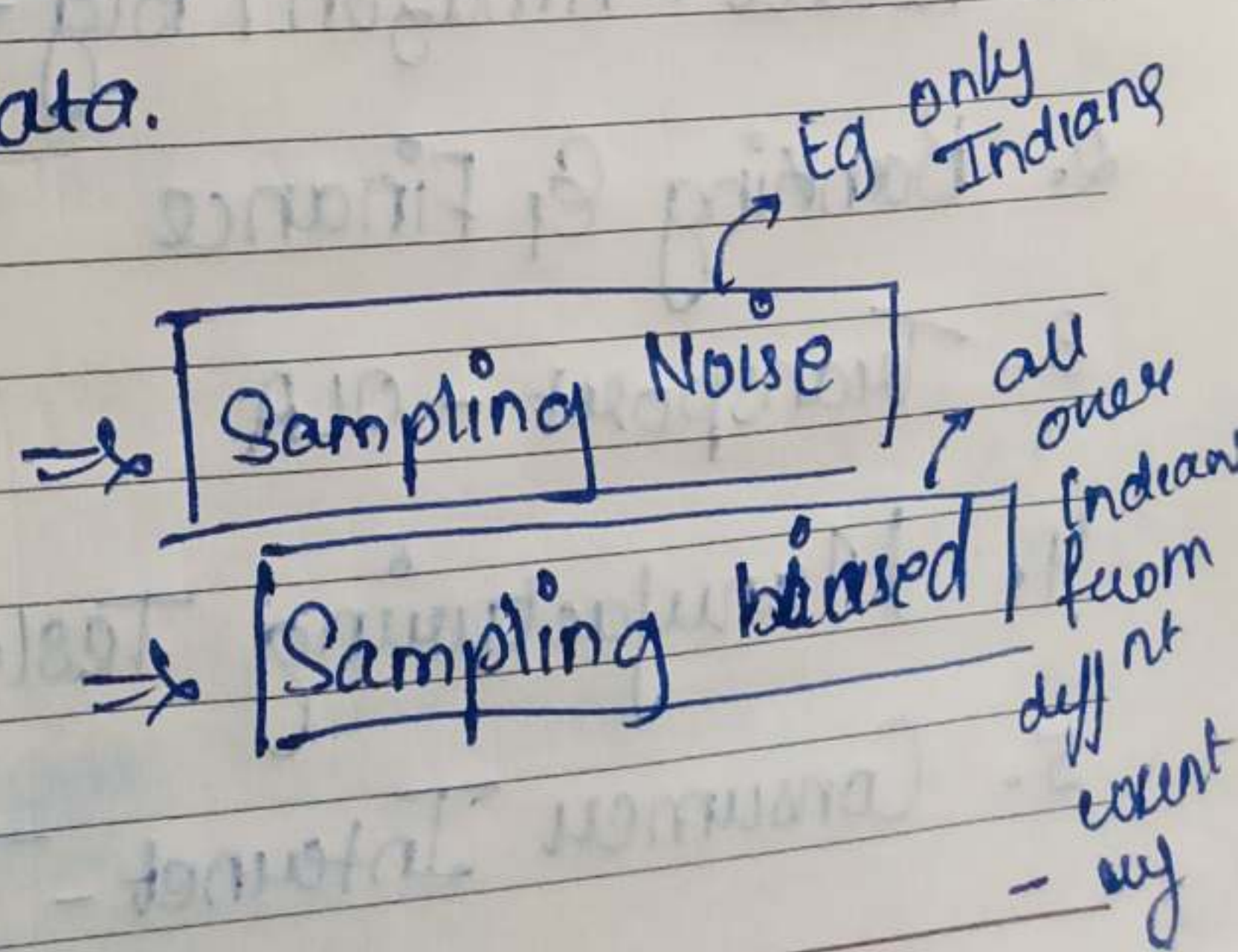
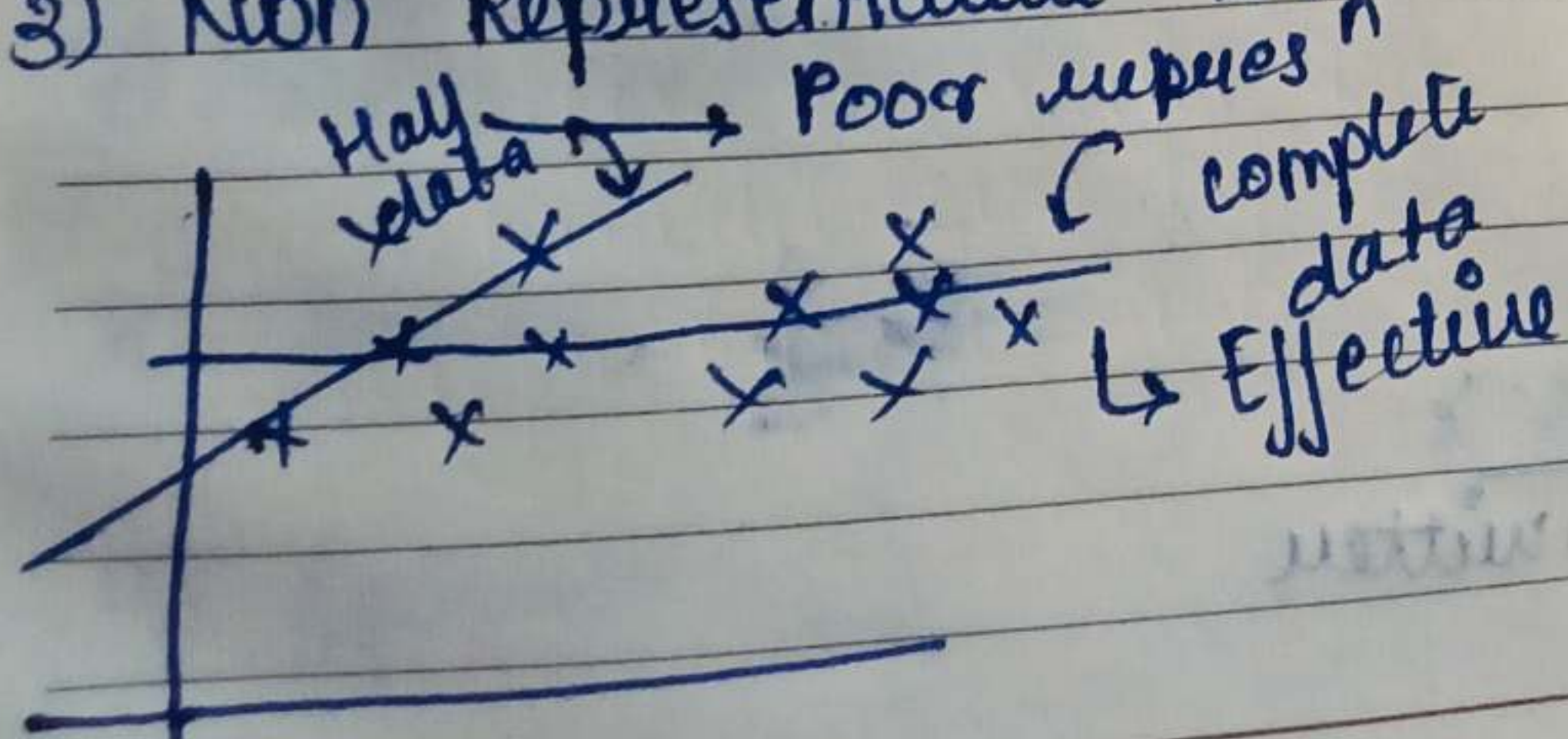
Challenges in Machine Learning

1) Data Collections



2) Insufficient data / Labelled data.

3) Non Representative Data



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23 24 25 26 27 28 29 30

Predict for unseen
incoming instance
using model

Predict for unseen
incoming instance using
training data directly

Can throw away
input/training data
after model training

Input/training data
must be kept since
each query uses part or
full set of ~~training~~ training
observation

Requires a known model
form

May not have explicit
model form.

Storing model generally
requires less storage

Storing training data generally
requires more storage

L-5

Offline Learning
Less complex as
model is constant

Features
complexity

Online Learning

Dynamic complexity as
the model keeps changing
evolving over time

Batch computations,
time batch-based
learning

Computational
Power

Continuous data
ingestions result in
consequent model refinement
computations

Easy to
implement

Use in
Production

Difficult to implement
and manage

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Tensor
Keras

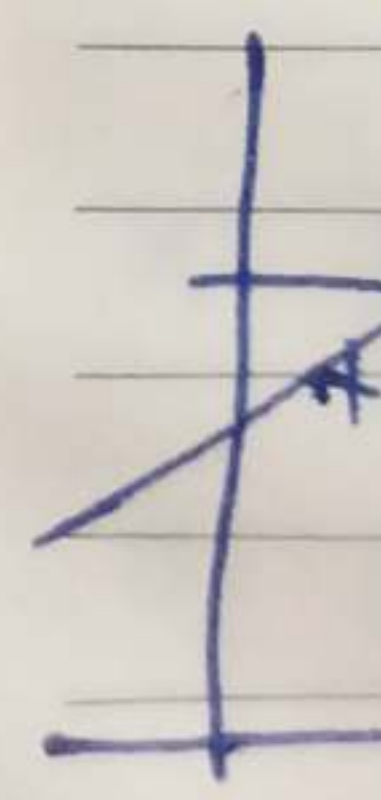
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Cho

1) Do

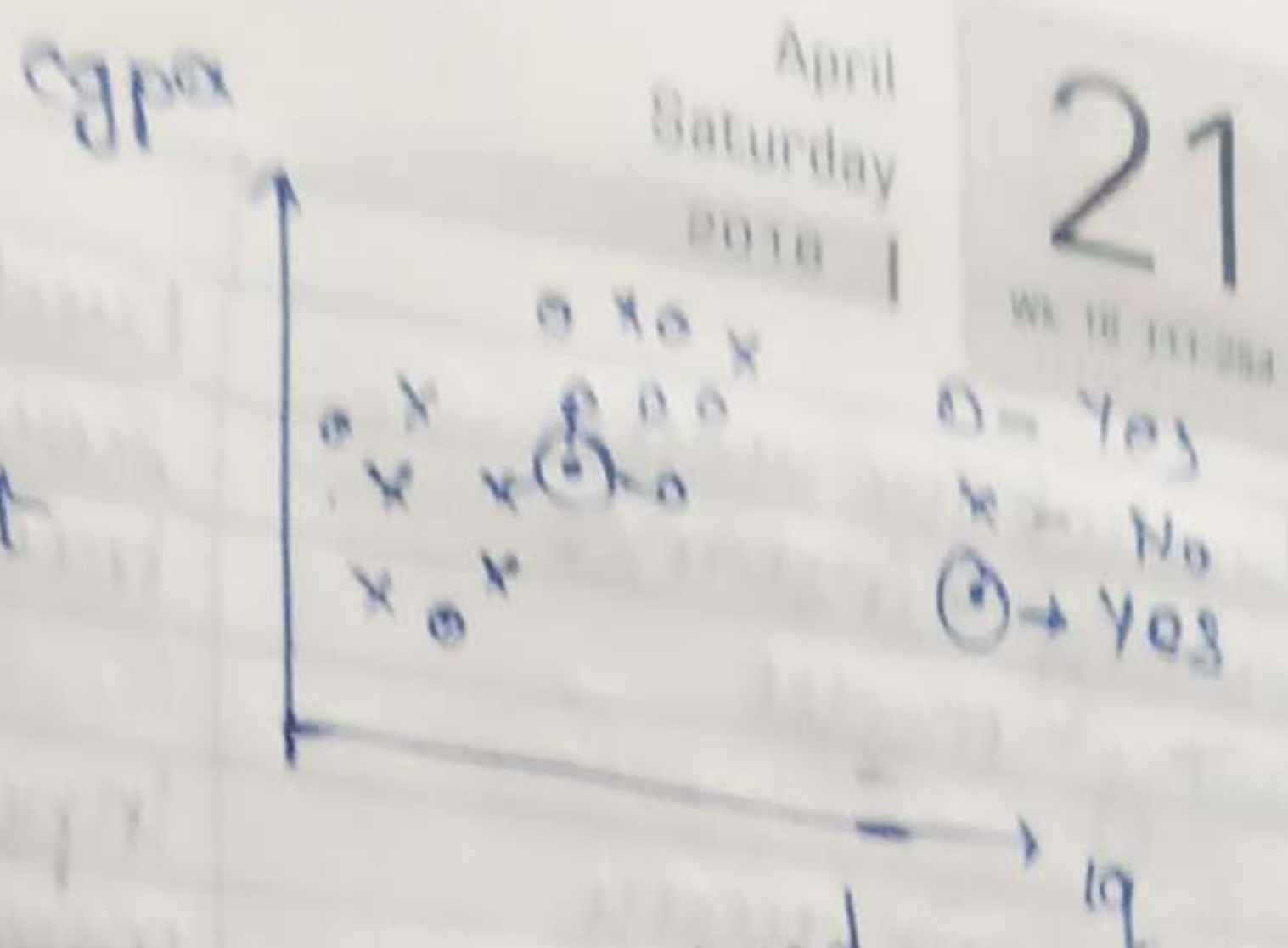
2) In

3) No



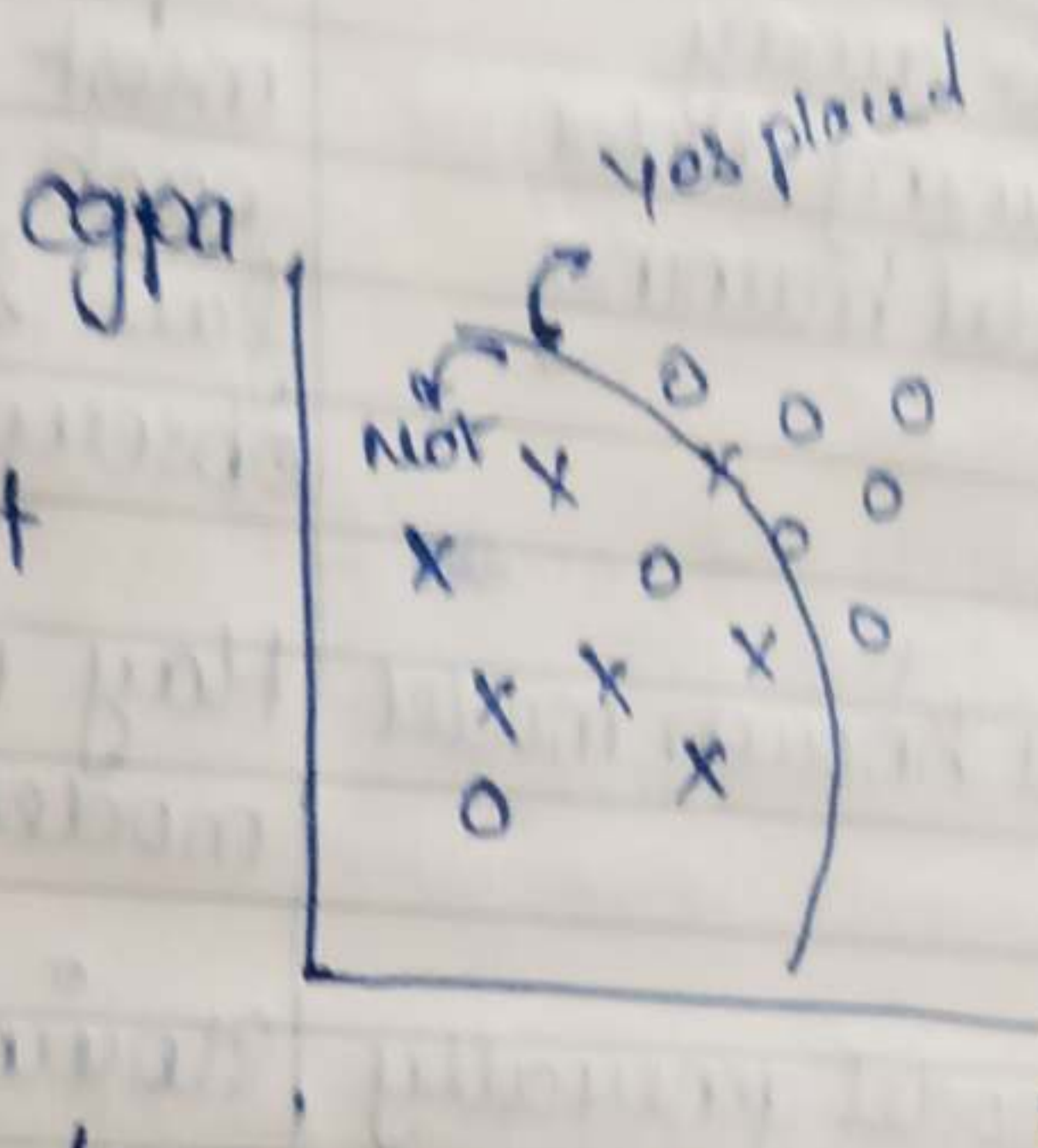
Instance Based

IQ	cgpa	placement
80	8	Y
70	7	N
108	7.5	



Modal Based learning

IQ	cgpa	placement
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Usual /conventional ML

- Prepare the data for model Training
- Train model from training data to estimate model parameters i.e. discover pattern
- Store the model in suitable form
- Generalise the rules in form of model, even before scoring instance is seen

Instance Based Learning

- Prepare the data for model training
- Do not train model. Pattern discovery postponed until scoring query received
- There is no model to store
- No generalization before scoring. Only generalise for each instance individually as and when seen

L-5

Online Machine Learning

↳ Incremental learning using mini batches of data where you train your data sequentially

Q. When to use

- 1) where there is a concept drift
- 2) Cost Effective
- 3) Faster solution.

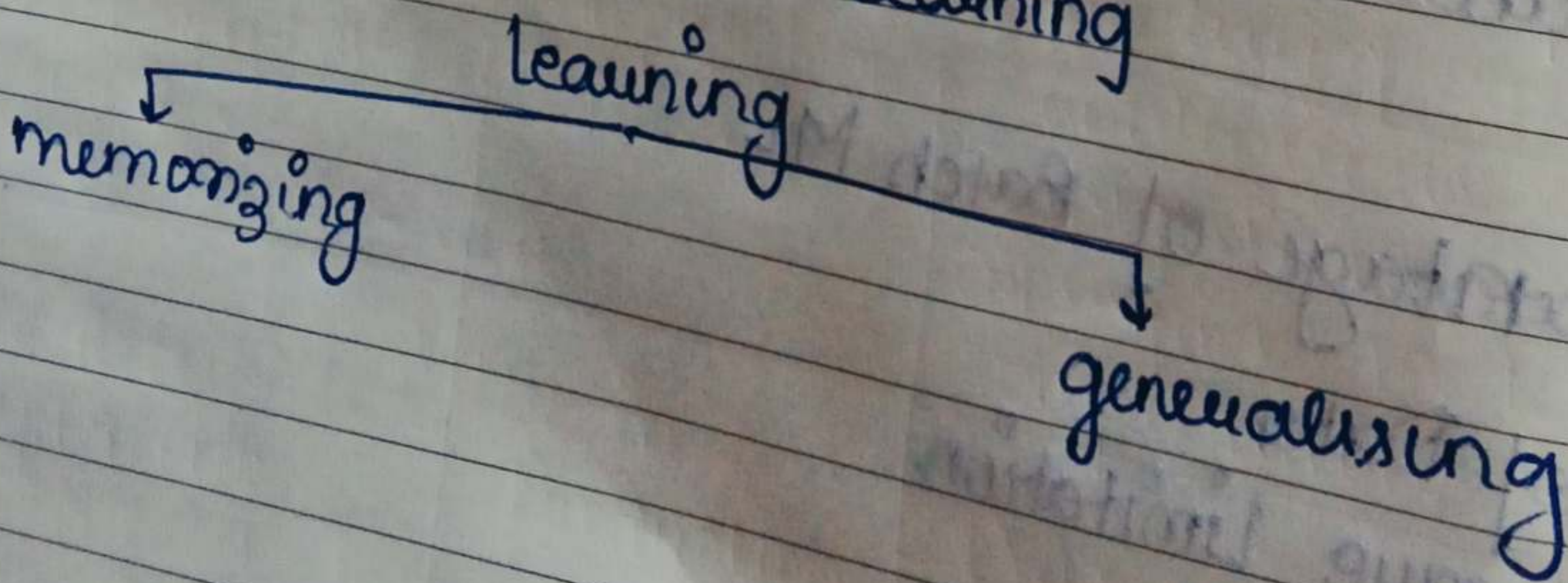
How to implement

Learning rate → coding

Disadvantage → 1) Tendency to overfit
2) Risky

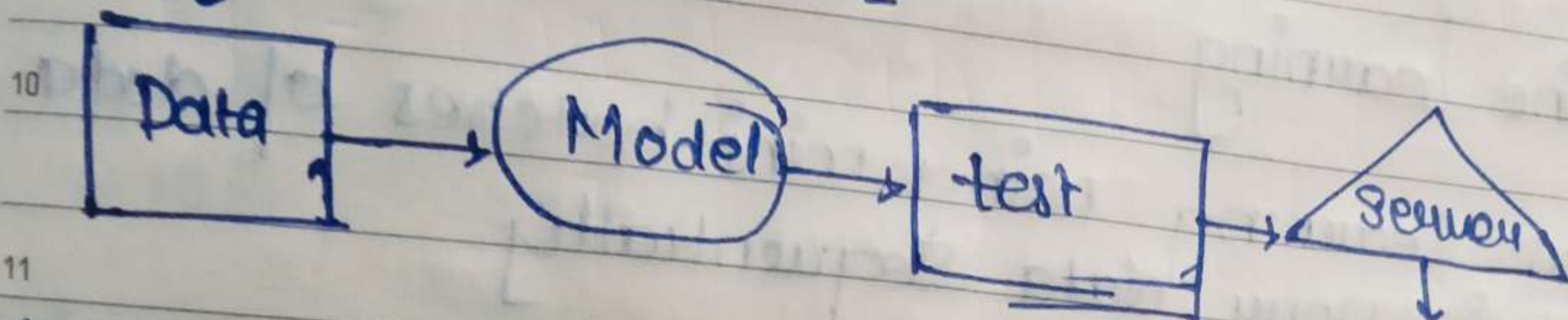
L-6

Instance Vs Model Based Learning



08 L-4

09 Batch Vs Online ML



11 * The problem with batch learning
12 ↳ It does not evolve / We have to retrain
01 this model.

02 Batch learning

03 ↳ Is where you train your ML model with your
entire data there is no incremental learning
04 & that is why since generally data is
big you don't train it ^{online} only instead you pull
05 down your model and re-train it with new
06 data and then we put it on server again &
again

07 * Disadvantage of Batch ML

- 1) Lots of Data
- 2) Hardware Limitation
- 3) Availability.

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23 24 25 26 27 28 29

* Anomaly detection

↳ Sort of outliers or when something is out of group of datas.

* Association Rule learning

↳ mining data and then make useful prediction

* Semisupervised

↳ Labeling with just a single data

Reinforcement learning → No data

- 1) Observe
- 2) Select action using policy
- 3) Action
- 4) Get reward or penalty
- 5) Update policy (learning)
- 6) Iterate

optimal policy found.

* Regression → If output data has both input & output and my output is numerical then it is regression

Eg: -

Iq	CGPA	Package
8.7	8.9	6
111	8.9	4

Regression

* Classification → have categorical values.

2) Unsupervised ML

↳ Only Input

Eg →

Iq	cgpa
70	7.0
60	6.0
100	9.

* Clustering → grouping

* Dimensionality reduction

Eg → lots of column → 1 column

No. of rooms | No. of washroom → sq ft → Area

↳ Feature extraction

* PCA

Visualisation →